UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	INTERACTIVE MULTIMEDIA
COURSE CODE	:	CSC413
EXAMINATION	:	NOV 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC413 mock paper (NOV 2026).
Answers show the expected marking points; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 15 x 1 mark; Part B = 15 x 1 mark; Part C = 30 x 1 mark; Part D = 4 x 10 marks; total = 100.

4. Questions: 64. Slots: 72. Placeholders: 0. No LLM call.
Questions and answers are for local study use only.

5. Structure derived from 15 unique CSC413 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

© Hak Cipta Universiti Teknologi MARA

PART A - ANSWER KEY

Q	Ans	Statement	Explanation
A1	TRUE	RGB is an additive colour model.	RGB adds red, green and blue light.
A2	TRUE	Chroma key compositing replaces a colour background with another image.	Chroma key removes a selected colour.
A3	FALSE	Higher resolution always means a smaller file.	Higher resolution usually increases file size.
A4	TRUE	A scanner is an input device.	A scanner captures documents into the computer.
A5	TRUE	A web browser displays hypertext documents.	Browsers render web pages.
A6	TRUE	HTML is used to structure content on web pages.	HTML marks up web content.
A7	TRUE	Card-based authoring tools are best for messages with a clear beginning and end.	Cards support sequential navigation.
A8	TRUE	CMYK is used for printed output.	CMYK is a subtractive print model.
A9	TRUE	Multimedia combines text, graphics, audio, video and animation.	These are the core multimedia elements.
A10	FALSE	3D animation does not use modelling.	3D animation relies on models.
A11	TRUE	Keyframes define important positions in an animation.	Keyframes mark major changes.
A12	TRUE	A bitmap image is made up of pixels.	Bitmap images use a pixel grid.
A13	FALSE	A linear presentation allows users to choose their own path.	A linear presentation follows a fixed sequence.
A14	TRUE	A prototype is an early model of a product.	Prototypes test ideas before production.
A15	FALSE	Immersion is not important in virtual reality.	Immersion is central to VR.

© Hak Cipta Universiti Teknologi MARA

PART B - ANSWER KEY

Q	Ans	Term	Definition
B1	J	Browser	software that displays web pages
B2	O	Keyframe	important frame in an animation
B3	C	Scanner	input device that digitises documents
B4	N	Tweening	generating in-between frames
B5	E	Frame rate	frames displayed per second
B6	G	Cel animation	hand-drawn frame-by-frame animation
B7	K	Storyboard	visual sequence plan
B8	A	Card-based tool	authoring tool organised like pages or cards
B9	L	Plotter	output device for large drawings
B10	D	Chroma key	replacing a colour background
B11	F	Copyright	exclusive rights of a creator
B12	B	Serif	font with small strokes at character ends
B13	I	HTML	language for web structure
B14	H	Bitmap	image made of pixels
B15	M	Multimedia	integration of text, graphics, audio, video and animation

© Hak Cipta Universiti Teknologi MARA

PART C - ANSWER KEY

Q	Ans	Correct answer
C1	C	Scanner
C2	C	Loudness
C3	C	Intended users
C4	D	Frequency
C5	A	Pixels
C6	A	A cinema presentation
C7	C	Draws alternate lines of a frame
C8	D	Temporary working memory
C9	A	WAV
C10	D	Replace a colour background
C11	C	Screen display
C12	D	Space between individual character pairs
C13	A	Card-based
C14	C	An early model of a product
C15	B	Multimedia with hyperlinks
C16	A	Text, graphics, audio, video and animation
C17	A	Number of pixels in an image
C18	B	Analog sound to digital values
C19	C	Small strokes at character ends
C20	B	Improves accuracy
C21	B	Along a predefined route
C22	A	Compiler
C23	A	Musical notes and instructions
C24	A	Play while downloading
C25	C	The user can control content and flow
C26	B	Non-linear
C27	B	Plotter
C28	C	A visual sequence plan
C29	A	It combines visuals and sound
C30	D	Important frames in an animation

© Hak Cipta Universiti Teknologi MARA

PART D - WORKED ANSWERS

QUESTION 1

Topic: authoring interactivity

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

Interactivity support and export formats are important features.

(3 marks)

b) Explain the interactivity level suitable for the demonstration.

A medium level with menus and buttons lets users explore without overwhelming them.

(3 marks)

c) Explain TWO authoring tool categories appropriate for this project.

Card-based tools suit structured screens; timeline-based tools suit animation and media sequencing.

(4 marks)

QUESTION 2

Topic: web hypermedia

A museum wants a website that presents its collection and lets visitors explore related exhibits.

a) State TWO differences between linear and non-linear navigation.

Linear follows a fixed sequence; non-linear lets users choose their own path.

(3 marks)

b) Describe TWO roles of hyperlinks in the website.

They connect related exhibits and let users move between pages.

(3 marks)

c) Explain how streaming and compression affect multimedia delivery on the web.

Compression reduces file size for faster transfer; streaming lets audio or video play while it downloads.

(4 marks)

QUESTION 3

Topic: graphics

A product catalogue must show sharp images on screen and in print.

a) State TWO differences between bitmap and vector images.

Bitmap images use pixels and can pixelate; vector images use mathematical objects and scale cleanly.

(2 marks)

b) Describe TWO factors that affect image quality.

Resolution and colour depth both affect how much detail an image can show.

(3 marks)

c) Explain how colour model and resolution affect screen and print output.

Screens use RGB and print uses CMYK; print normally needs a higher resolution for sharp output.

(5 marks)

© Hak Cipta Universiti Teknologi MARA

QUESTION 4

Topic: animation

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

GIF and MP4 are suitable; SWF is also acceptable.

(2 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

Draw each frame, capture the frames in sequence, then play them back; it gives direct artistic control.

(4 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

2-D uses flat drawings such as a cartoon; 2.5-D adds depth to flat artwork, such as parallax layers in a game.

(4 marks)

END OF MARKING SCHEME

© Hak Cipta Universiti Teknologi MARA